Convex and Quasi-convex Functions

Convex Optimization, Homework 3

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English summary of my original Persian report, which is kept as a PDF in this folder.

This homework is about deciding whether functions are convex, concave or quasi-convex, using second-order conditions and the composition rules.

Problems and what I did

1. **Running average of a function.** I showed by differentiation that $g(x) = \frac{1}{x}\int_0^x f(t)\,dt$ inherits monotonicity and concavity from f.
2. **Epigraphs.** I matched the epigraph types to the functions: a half-space comes from an affine function, a convex cone from a positively homogeneous function, and a polyhedron from a piecewise-linear function.
3. **Hessian tests.** For each function I computed the Hessian. When it was indefinite, I checked the sub- and superlevel sets instead: some functions are quasi-concave only, some quasi-linear (both quasi-convex and quasi-concave), and some convex and concave as well as quasi-convex.
4. **A function of several variables.** I computed the Hessian entries for $i \neq j$ and $i = j$, and used an inequality between two auxiliary vectors a and b to show that the function is concave.
5. **Composition rules.** I used them for
   - the sum of the k largest residuals of $Ax - b$, a pointwise maximum of convex functions;
   - $-\log(-g(x))$ for convex $g < 0$, where $-\log$ is convex and decreasing;
   - $y^T y/t$ composed with an affine map, a perspective of a quadratic;
   - a log of a sum of exponentials, through Jensen's inequality.
6. **Products and ratios.** Using Jensen's inequality, I showed that the product of two non-negative convex functions that are both non-decreasing (or both non-increasing) is convex, derived the matching results for concave functions, and handled f/g through $1/g$.
7. **More compositions.**
   - $\|Ax - b\|$ is convex as a norm of an affine map.
   - $\|Ax + b\|/(c^T x + d)$ is convex on $c^T x + d > 0$ as the perspective of a norm.
   - $\max\{1, x, x^2\}$ is convex as a pointwise maximum.
   - A function whose second derivative changes sign at $x = -2$ is convex for $x > -2$ and concave for $x < -2$.
8. **Perspective-based functions.**
   - $t - \frac{1}{t}x^T x$ is concave.
   - $-\log(t - x^T x/t)$ is convex.
   - A sum of two negated logarithms of concave functions is convex.
9. **Linear-fractional functions.** I showed that $u(x)/v(x)$ with affine u and positive affine v is quasi-convex, since each sublevel set is a half-space. Whether it is actually convex depends on the sign of a remaining term.

10. **A three-variable function.** $(x-z)^2/(y+1)+\max\{1+2x-y,\ 1/z,\ x^2\}$ is convex on $y > -1$, $z > 0$. The first term is a quadratic-over-linear, and the second is a pointwise maximum of convex functions.